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Economic evaluation of the benefits of reducing acute cardiorespiratory morbidity associated with air pollution

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Abstract

Background: Few assessments of the costs and benefits of reducing acute cardiorespiratory morbidity related to air pollution have employed a comprehensive, explicit approach to capturing the full societal value of reduced morbidity.

Methods: We used empirical data on the duration and severity of episodes of cardiorespiratory disease as inputs to complementary models of cost of treatment, lost productivity, and willingness to pay to avoid acute cardiorespiratory morbidity outcomes linked to air pollution in epidemiological studies. A Monte Carlo estimation procedure was utilized to propagate uncertainty in key inputs and model parameters.

Results: Valuation estimates ranged from \$13 (1997, Canadian) (95% confidence interval, \$0–28) for avoidance of an acute respiratory symptom day to \$5,200 (\$4,000–\$6,400) for avoidance of a cardiac hospital admission. Cost of treatment accounted for the majority of the overall value of cardiac and respiratory hospital admissions as well as cardiac emergency department visits, while lost productivity generally represented a small proportion of overall value. Valuation estimates for days of restricted activity, asthma symptoms and acute respiratory symptoms were sensitive to alternative assumptions about level of activity restriction. As an example of the application of these values, we estimated that the observed decrease in particulate sulfate concentrations in Toronto between 1984 and 1999 resulted in annual benefits of \$1.4 million (95% confidence interval \$0.91–1.8 million) in relation to reduced emergency department visits and hospital admissions for cardiorespiratory disease.

Conclusion: Our approach to estimating the value of avoiding a range of acute morbidity effects of air pollution addresses a number of limitations of the current literature, and is applicable to future assessments of the benefits of improving air quality.

Background

Numerous assessments have been conducted around the

world of the costs and/or benefits of interventions to improve air quality [1–7]. Benefits assessments typically

employ the damage function approach, consisting of estimating changes in air pollution emissions and resultant changes in ambient air quality; calculating the number of avoided adverse health and environmental outcomes using concentration-response functions; and applying economic value measures to avoided outcomes [8], often extrapolating valuation estimates from the original context in which data were collected [9]. The majority of benefits generally derive from improved human health, including both reduced mortality and morbidity. Benefits of reduced mortality and chronic morbidity tend to dominate these assessments, accounting for over 90 percent of benefits in most instances [1,7]. Nonetheless, acute morbidity is still viewed as an important component of the broad public health burden associated with air pollution [10], and benefits assessments will undoubtedly continue to include acute morbidity effects. Thus, future assessments will need to address deficiencies in past work dealing with acute morbidity, including failure to consistently employ a comprehensive, explicit approach to capturing all components of the societal value of reduced morbidity and differing levels of severity of symptoms and activity restrictions.

In earlier analyses [3,5], consideration of morbidity effects of air pollution was restricted to a limited range of symptoms and activity restrictions as observed in clinical and epidemiological studies. These assessments, as well as more recent ones [2], relied heavily on contingent valuation studies [11–13] in which individual willingness to pay (WTP) to avoid a variety of symptoms of varying duration was evaluated. In more recent assessments, there has been a greater reliance on cost of illness (COI) data, comprising medical costs and lost production experienced by society as a result of a health effect. However, these measures represent at best a lower bound on the total societal value of avoiding these health outcomes [14], since they reflect only one element of this value. In some assessments, no adjustments have been made to correct for this bias [6,7], while in others, empirical evidence from studies which simultaneously measured COI and individual WTP have been used to derive adjustment factors [1,4,15–17]. Such adjustments recognize deficiencies in available estimates, but applying them uniformly to a variety of cardiorespiratory conditions could introduce error in benefits assessments. An additional issue is that most previous assessments have relied heavily upon valuation estimates derived from U.S. data, the applicability of which outside the U.S. can be questioned.

In the present study, our objective was to estimate the benefits of avoiding a variety of acute cardiorespiratory morbidity outcomes which have been linked epidemiologically to air pollution, using complementary data on COI and individual WTP. In contrast to the lack of a con-

sistent, comprehensive approach in earlier studies, particular features of our approach include application to all outcomes of: a consistent conceptual model of the nature of acute episodes of cardiorespiratory disease, including differing levels of severity of symptoms and activity restrictions; a comprehensive valuation framework encompassing all aspects of the value of avoided morbidity and a recognition of the perspective (societal versus individual) from which primary data were collected; a common source of inputs based on empirical data on the duration and severity of episodes of cardiorespiratory disease; and an estimation procedure which propagates uncertainty in key inputs.

Methods

Conceptual model of acute cardiorespiratory morbidity

We identified several types of acute cardiorespiratory morbidity which have been linked with air pollution in epidemiological studies: respiratory and cardiac hospital admissions and emergency department visits, restricted activity days, asthma symptom days, and acute respiratory symptom days [18–21]. Based on this literature, we identified asthma, chronic obstructive pulmonary disease (COPD), respiratory infections, congestive heart failure, cardiac dysrhythmias, and myocardial infarction/angina, as well as mild non-specific respiratory symptoms, as the principal conditions of interest [18–22].

As a general framework, we viewed these conditions as occurring in the form of acute episodes superimposed upon the underlying baseline health state. The extent to which health status departed from its premorbid level could be expected to vary during the course of the episode, so that the episode of illness could be characterized as the sum of several periods, each of which is associated with a given decrement from baseline health status. Depending on the nature of the morbidity outcome, these periods might include periods spent in or out of hospital, and in various states of restriction of activity.

Valuation framework

Reduced morbidity due to cleaner ambient air represents real economic value to individuals and society, measured most appropriately in a welfare economic framework by WTP to improve health or willingness to accept compensation (WTA) to worsen it [23]. Measures such as quality- or disability-adjusted life years (QALYs or DALYs) represent an alternative measure which generically reflects preferences among health states involving loss of quality and/or quantity of life [24]. However, experience with application to acute effects of air pollution is limited [25], and unlike dollar valuation of health outcomes, these measures do not permit direct comparisons to costs. WTP (and WTA) are generally regarded as comprising several components: expenditures related to medical treatment of

adverse health effects, including costs of treatment as well as related out of pocket expenditures (for example, for assistance in completing usual unpaid chores that the individual is unable to undertake as a result of ill health); the dollar value of interruption in work duties, which can range from decreased productivity to absence from the workplace; expenditures to avert or mitigate future occurrences of the effect (eg. purchasing an air conditioning unit as a means of reducing the chances of an asthma attack associated with outdoor air pollution); and the dollar value of pain and suffering, inconvenience, restrictions and reduced enjoyment of leisure activities, anxiety about the future, and concern and inconvenience to family members and others [14]. The perspective of the analysis (societal versus individual) is clearly an important consideration, in that individual WTP will not reflect medical expenditures and lost productivity where there is third party payment of medical expenditures or paid sick leave [8,14,26].

A comprehensive measure of the total value, from a societal perspective, of avoiding the acute cardio-respiratory morbidity effects of air pollution can therefore be expressed as follows:

$$V_T = V_{AE} + V_{PS} + V_{COT} + V_{LP} \quad (1)$$

where,

V_{AE} = value of reduced averting expenditures;

V_{PS} = value of reduced pain and suffering;

V_{COT} = cost of treatment including reduced expenditures on mitigation of the illness; and

V_{LP} = value of lost productivity.

Sources of empirical data

We used empirical results from two recent studies in order to generate estimates of V_T . Both studies have been described in detail elsewhere [27–29], but are summarized briefly below.

In the first study, joint estimates were constructed of individual WTP to reduce acute episodes of respiratory and cardiovascular ill health based on graded-pairs and choice-based stated preference survey responses among 399 randomly selected subjects in Toronto in 1997 [28,30]. The survey instrument employed health-state descriptions based on the Quality of Well Being (QWB) index [31,32], with attributes comprising symptoms, motor function, and ability to fulfill roles and expectations. The scope of symptoms was selected to correspond to the cardiorespiratory conditions of policy interest as described

above. Four levels of activity restriction were considered (Table 1). The payment vehicle employed in this survey was described as illness related costs not covered by the government health system or a company insurance plan. Because pre-testing revealed that respondents with and without paid sick leave interpreted costs differently and this led to ambiguity in interpretation of their WTP, respondents were instructed to assume that they had paid sick leave. By combining results from graded-pairs and choice formats, the resulting estimates from this study are more robust than estimates based on a single format. In addition, relative to other studies in the WTP literature in this area, estimates from this study were based on comprehensive multi-attribute health state descriptions which ensured that key attributes were clearly and consistently presented to respondents, for the wide range of symptoms and severity levels covered in the study. Estimates from this study were not systematically smaller or larger than other literature and varied logically with duration and severity of outcomes. However, given the payment vehicle and the instructions regarding sick leave, we determined that individual WTP estimated in this study would not reflect V_{LP} or that portion of V_{COT} covered by public health care system or third party insurance. We obtained these data from the following study.

In the second study, data were collected prospectively on individuals visiting emergency departments for cardiorespiratory conditions, as part of an epidemiological study of air pollution in Saint John, Canada between 1992 and 1996 [29,33]. A random sample of those agreeing to follow-up was contacted two weeks after their visit and asked to complete a structured telephone interview, in which information was collected on the duration of the disease episode which prompted their visit, disposition of their visit (whether they were admitted or discharged), and the occurrence of various outcomes including restricted activity days, days of missed work for the patient, parent, or other care giver, health care utilization and illness-related expenses, from the onset of their earliest symptoms until the follow-up interview. Data were obtained for 1,772 individuals. Inpatient utilization data were also abstracted from the clinical records of 393 randomly selected patients representing the same groups of cardiorespiratory conditions, and cost of illness estimates were generated which included fully allocated costs related to hospital utilization and emergency department visits, as well as costs of physician visits, medication use, equipment, and out-of-pocket expenses [27]. We analyzed these data using stepwise ordinary least squares regression. Selection of the final multi-variate model was based on minimization of Akaike's Information Criterion, the deviance penalized for the number of parameters being estimated [34]. Compared to other literature in this area, this study has the advantage of having been designed specifically to

Table 1: Description of symptoms and activity restrictions

Type of symptom/activity restriction	Description
Upper Respiratory	Stuffy or runny nose and sore throat.
Eye Irritation	Eye irritation.
Asthma/COPD ^a	Coughing, wheezing, shortness of breath.
Respiratory Infection	Coughing or wheezing with fever, chills, aching all over.
Dysrhythmia	Fluttering in chest and feeling light-headed
Congestive Heart Failure	Shortness of breath, and swelling in ankles and feet.
Myocardial Infarction/Angina	Pain in chest or arm.
Mild Limitations ^b	Can go to work, go to school, do housework, and participate in social or recreational activities, but have some physical limitations (trouble bending, stooping, or doing vigorous activities because of this health condition).
	Can go to work, go to school, do housework, but have some physical limitations (trouble bending, stooping, or doing vigorous activities), and cannot participate in social or recreational activities because of this health condition.
Housebound	Cannot leave house, go to work, go to school, do housework, participate in social or recreational activities, and have some physical limitations (trouble bending, stooping, or doing vigorous activities) because of this health condition, but can care for self.
Help caring for Self	Cannot leave house, go to work, go to school, do housework, participate in social or recreational activities, and need help caring for self (feeding, bathing, dressing, toilet).
In Hospital	In hospital and need help caring for self (feeding, bathing, dressing, toilet)

^achronic obstructive pulmonary disease ^bcollapsed into a single level corresponding to mild restrictions, for the purposes of the final model used to generate WTP estimates.

prospectively quantify the broad quality of life impacts and COI associated with episodes of cardiorespiratory disease, thus ensuring that the required data elements were collected in a consistent, coordinated fashion. Cost analysis was carried out in a manner consistent with current methodological standards.

Estimation procedure

Based on these two studies, we calculated V_T according to Equation (1) as follows:

$$V_{PS} + V_{AE} = \text{value from stated preference survey} = V_{SP}(S_i A_{ij} D_k) \quad [28,30]$$

where S_i , A_{ij} , and D_k are the symptom complex, level of activity restriction and duration of the illness episode respectively (See [28] for details of the stated preference valuation estimation procedure).

$V_{COT} = f(\text{diagnosis, duration, admission to critical care/non critical care})$ (See results section below for model specification);

$$V_{LP} = D_{LP} \times W_D$$

where D_{LP} = duration of lost productivity in days; and W_D is the average daily wage for Canada in 1997 of \$119.60 [35].

While recent recommendations note the friction cost method as an alternative to the conventional human capital approach as potentially more accurately reflecting the ability to replace disabled workers, thus minimizing societal productivity losses [26,36], we assumed that this was not applicable in the current context, where illness episodes are of short duration. We did not attempt to value productivity losses not resulting in lost time (i.e. reduced work capacity), but we did capture time lost by parents and other care givers.

In estimating V_{SP} , we mapped symptoms and activity restrictions to the various health outcomes identified in epidemiological and clinical studies as shown in Table 2. Some outcomes were mapped to time periods both in and out of hospital, in keeping with the expected natural history of an acute episode of cardiorespiratory morbidity. Hospital admissions, for example, were considered to occur following a period over which symptoms and activity restrictions began and worsened to the point that hospital admission was required, and in turn, to be followed by a period of recovery out of hospital before returning to the baseline health state. Similarly, a proportion of emergency department visits was expected to result in eventual admission to hospital, with accompanying periods of symptom onset and recovery. While, *a priori*, both asthma symptom days and restricted activity days would include days in hospital, on the basis of population-based data, we estimated that the latter would account for a very small